

Time and Causality

Chapter 5

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Data Science and Analytics

- 1 Time Series Analysis
 - ARMA and ARIMA Models
 - Selecting Optimal Lag Length: AIC and BIC
- 2 Causal Analysis
- 3 Summary

Why do we need time series analysis?

- Stock market prediction
- Weather forecasting
- Energy demand
- Sales forecasting
- COVID-19 daily cases
- Sensor monitoring
- Traffic prediction

Example: Airline Demand Forecasting

Scenario

An airline has recently experienced declining profitability due to **underutilization**.

- Too many flights are scheduled.
- Too few passengers purchase tickets.
- Aircraft fly with many empty seats.
- Operating costs remain high while revenue decreases.

Business Problem

How many passengers are expected to travel on a particular route next month?

Solution

Time series analysis models historical passenger counts to identify patterns and forecast future demand.

Time Series

- sequential set of data points, measured typically over successive times
- Time series analysis comprises methods for analyzing such data to extract meaningful statistics and other characteristics
- To understand how data changes over time → help track past patterns and predict future values

Two different perspectives

- 1 **Time-Domain Analysis:** investigates relationships across time
- 2 **Frequency-Domain Analysis:** investigates repeating cycles, hidden periodicities, seasonal frequencies

Method

- Focuses on data collected at regular time intervals
- Helps identify trends, seasonality and sudden changes
- Useful for planning, prediction and decision-making
- Common methods: ARIMA, exponential smoothing and machine learning models

Time-Domain Analysis

- how the current value of a time series depends on its previous values
 - How does the past influence the future?

$$X_t = f(X_{t-1}, X_{t-2}, \dots) + \epsilon_t$$

- X_t = current observation
 - X_{t-1} = previous observation
 - ϵ_t = random error
- studies how past observations influence future observations
 - primary approach for forecasting
- concepts: autocorrelation, stationarity, ARMA, and ARIMA

Temporal Dependence Illustration

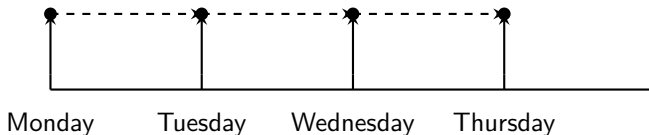
Example: Daily Temperatures

Day	Temperature
Monday	28° C
Tuesday	29° C
Wednesday	30° C

Observation: The temperature tomorrow is likely to be close to today's temperature.

$$X_t \text{ depends on } X_{t-1}$$

Time-Domain Illustration:



Concept: Past values influence future values: called **temporal dependence**

Frequency-Domain Analysis

- studies how much of the variation in a time series occurs at different frequencies
- Every time series can be represented as a combination of sine and cosine waves with different frequencies

$$X(t) = \sum_{k=1}^n A_k \sin(2\pi f_k t + \phi_k)$$

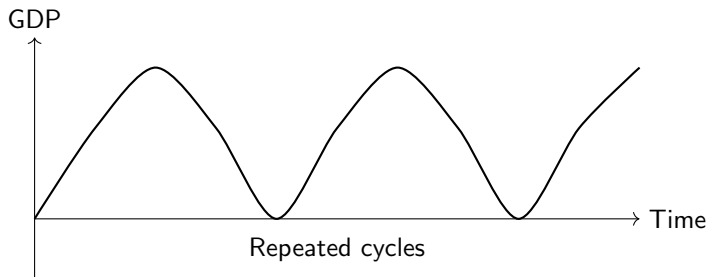
- A_k : amplitude (strength of cycle)
- f_k : frequency (cycles per unit time)
- ϕ_k : phase angle (shift)

Example: Economic Cycles

Scenario: Suppose a country's GDP rises and falls every 8 years.

- **Time-domain question:** How does last year's GDP affect this year's GDP?
- **Frequency-domain question:** Is there an 8-year economic cycle?

Illustration:



Frequency-Domain Methods

- Fourier Transform
- Fast Fourier Transform (FFT)
- Spectral Density Estimation
- Periodogram
- Wavelet Transform
- Short-Time Fourier Transform (STFT)

Applications of Frequency-Domain Analysis

- **Economics:** Business cycles, expansion, recession
- **Electrical Engineering:** Signal filtering, communication systems, radar
- **Healthcare:** ECG, EEG, brain wave detection
- **Climate Science:** Rainfall cycles, El Niño, temperature oscillations
- **Astronomy:** Planetary motion, sunspot cycles

Time vs. Frequency Domain

Feature	Time Domain	Frequency Domain
Focus	Sequential observations	Cycles and periodicities
Question	How does the past affect the future?	What repeating patterns exist?
Independent Variable	Time	Frequency
Main Tool	Autocorrelation	Fourier Transform
Models	AR, MA, ARMA, ARIMA	FFT, Spectral Analysis
Applications	Forecasting	Signal processing
Unit	Lag	Hertz (cycles per unit time)

Components of a Time Series

A time series is generally affected by four components:

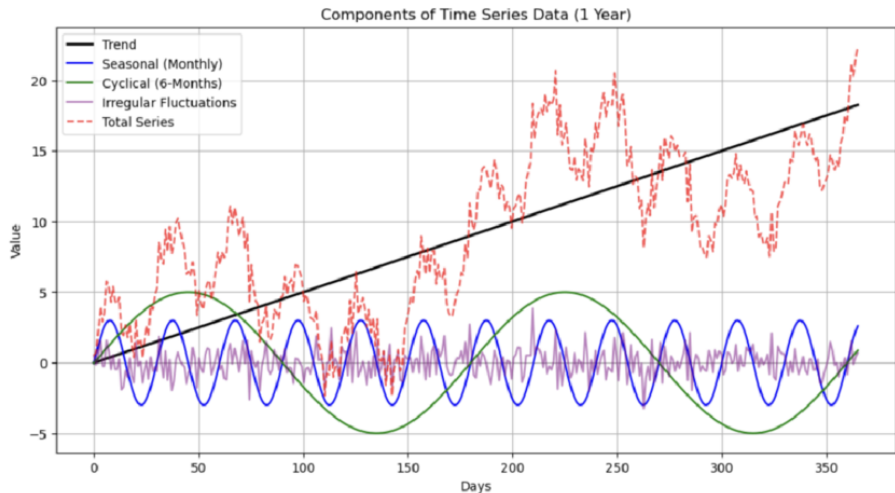
- **Trend (T_t):** the general tendency to increase, decrease, or stagnate over a long period.
- **Seasonal variation (S_t):** fluctuations within a year tied to climate, customs, or calendar effects.
- **Cyclical variation (C_t):** medium-term changes that repeat in cycles, typically longer and less regular than seasonal effects.
- **Irregular variation (I_t):** unpredictable influences (war, strike, earthquake) with no defined statistical pattern.

Combining the Components

Additive model: $Y(t) = T(t) + S(t) + C(t) + I(t)$ (components independent)

Multiplicative model: $Y(t) = T(t) \times S(t) \times C(t) \times I(t)$ (components may interact)

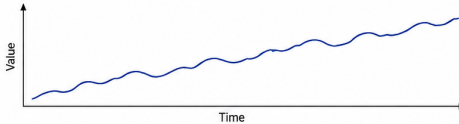
Components of a Time Series



Components of Time Series Data

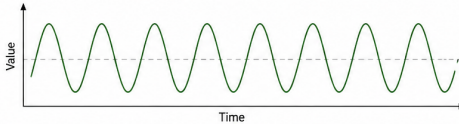
Trend (T_t)

Long-term movement in the data showing growth or decline.



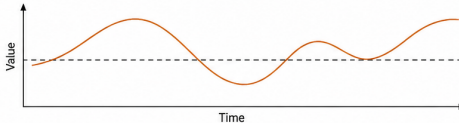
Seasonal variation (S_t)

Regular, repeating patterns that occur at fixed time intervals.



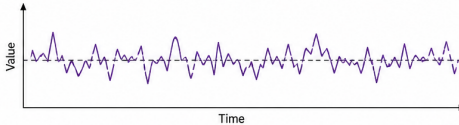
Cyclical variation (C_t)

Long-term fluctuations around the trend that are not of a fixed period (associated with business or economic cycles).



Irregular variation (I_t)

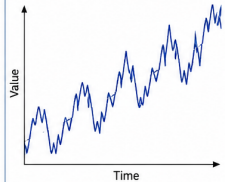
Random, unpredictable fluctuations due to unusual or unforeseen events (noise).



Combined Time Series

The observed time series is the combination of all components.

$$Y_t = T_t + S_t + C_t + I_t$$



T_t : Trend component
 S_t : Seasonal component
 C_t : Cyclical component
 I_t : Irregular component

Note: The components are not directly observed. They are conceptual parts used to understand the underlying structure of the time series.

Trend: long-term increase or decrease in a time series Examples

- Population growth
- Rising housing prices
- Increasing internet users
- Declining newspaper sales

Trend reflects the overall direction of the data.

Seasonality: refers to regular and predictable patterns that repeat over fixed intervals.

Examples

- Ice cream sales increase every summer.
- Retail sales increase every December.
- Electricity demand rises every evening.
- Tourist arrivals increase during holidays.

Cycles: long-term fluctuations that occur without a fixed period

Examples

- Business cycles
- Economic expansion
- Economic recession
- Housing market cycles

Unlike seasonality, cycles have no fixed duration.

Irregular variation consists of random, unpredictable fluctuations.

Examples

- Natural disasters
- Pandemic outbreaks
- Machine failure
- Political events
- Unexpected weather

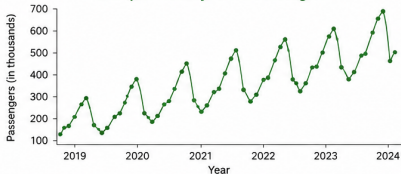
This component is also called **noise** or the **residual**

Seasonal vs Non-Seasonal Data

Seasonal Data

Data that exhibit regular, repeating patterns or cycles at fixed time intervals.

Example: Monthly Airline Passengers



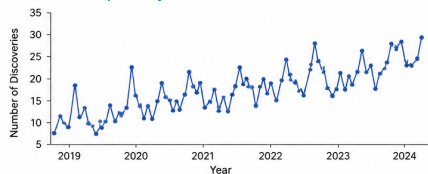
Characteristics

- Clear seasonal pattern (e.g., yearly, quarterly, weekly)
- Has trend, seasonality, and irregular components
- Example periodicity: 12 months ($s = 12$) for monthly data

Non-Seasonal Data

Data that do not exhibit regular repeating patterns at fixed time intervals.

Example: Daily Number of Scientific Discoveries



Characteristics

- No clear repeating seasonal pattern
- Usually only trend and irregular components
- No fixed periodicity

Aspect	Seasonal Data	Non-Seasonal Data
Pattern	Repeats at regular intervals	Does not repeat
Components	Trend + Seasonality + Irregularity (+ Cycles)	Trend + Irregularity (+ Cycles)
Examples	Airline passengers, Sales, Electricity demand, Temperature, Rainfall	Stock prices, Scientific discoveries, Population growth, Bitcoin price
Modeling	Use Seasonal models (SARIMA, Holt-Winters)	Use Non-seasonal models (ARIMA, ARMA, etc.)

Key Takeaway: Identifying whether data is seasonal or non-seasonal is the first and most important step before building an appropriate time series model.

When to Use (and Avoid) Time Series Forecasting

Use Time Series Forecasting When:

- Historical data is consistent and reliable.
- Clear temporal patterns (trend, seasonality) exist.
- Future behavior depends heavily on past observations.

Avoid Time Series Forecasting When:

- Data is highly erratic (e.g., a pure random walk).
- No meaningful historical patterns are present.
- Key influencing factors are external, unknown, or unpredictable.

Types of Time Series: Structure & Interval

1. Univariate vs. Multivariate

Univariate: Records only *one* variable over time. Simpler to model.

Example: Daily closing price of a stock.

Multivariate: Tracks *multiple* related variables to capture interdependencies.

Example: Weather forecasting (temperature, humidity, wind).

2. Continuous vs. Discrete

Continuous: Observed at every moment or very high frequency.

Example: ECG heart signals, IoT sensor streams.

Discrete: Recorded at fixed, regular intervals (most common).

Example: Monthly retail sales, daily website traffic.

3. Stationary vs. Non-Stationary

Stationary:

- Constant mean, variance, and pattern over time.
- No trend or seasonality.
- **Best for:** ARMA models, basic linear regression.

Non-Stationary:

- Changing patterns (trends, seasonality).
- Mean or variance shifts over time.
- **Best for:** ARIMA (needs differencing), SARIMA, Holt-Winters.

Stochastic Process

A stochastic process is a collection of random variables indexed by time:

$$\{X_t, t \in T\}$$

where

- X_t = random variable at time t
- $T = \{1, 2, 3, \dots\}$ denotes discrete time
- Each observation is a realization of a random variable.
- The observed sequence is called a **sample path**.
- Successive observations may be statistically dependent.

$$x_1, x_2, x_3, \dots, x_n$$

is one realization of the stochastic process.

Time Series as a Stochastic Process

Stochastic Process

A time series is modeled as a collection of random variables indexed in time, $X_1, X_2, X_3, \dots$, with t typically discrete, $t = 0, \pm 1, \pm 2, \dots$. The collection $\{X_t\}$ is the **stochastic process**; the observed values are a **realization** of that process.

Mean Function

$$\mu_t = \mu_{X_t} = E[X_t] = \int_{-\infty}^{\infty} x f_t(x) dx$$

- For white noise: $\mu_{w_t} = E[w_t] = 0$ for all t .
- For random walk with drift ($\delta \neq 0$): $\mu_{X_t} = E[X_t] = \delta t$.

Time Series as a Stochastic Process

The observed data

100, 102, 101, 105, 107

is only one possible realization.

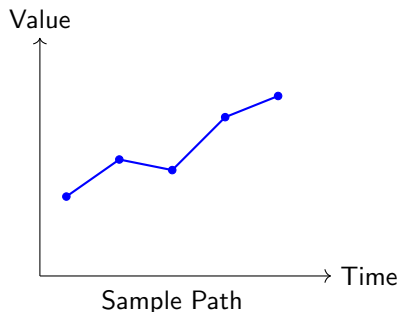
Future realizations may be

98, 101, 104, 108, 110

or

101, 99, 103, 102, 104

All are generated from the same stochastic process.



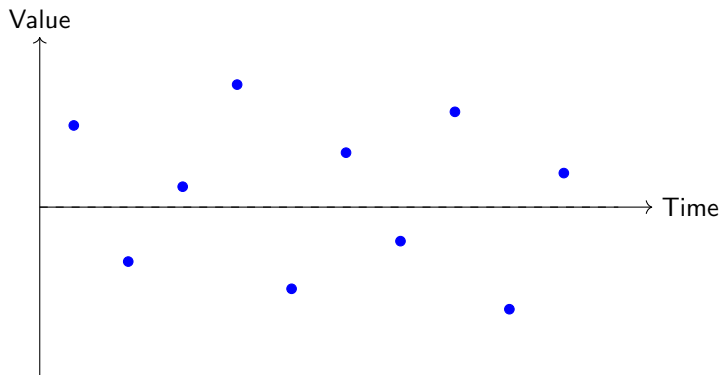
simplest stochastic process. Pure, unpredictable randomness

$$W_t \sim (0, \sigma^2)$$

Characteristics

- The simplest stochastic process. Pure, unpredictable randomness.
- **Properties:** Constant mean (0), constant variance (σ^2), and **zero autocorrelation**.
- **Crucial Rule:** The residuals (errors) of a good forecasting model must behave like white noise.

Illustration of White Noise



Random observations fluctuate around zero without any pattern.

Numerical Example of White Noise

Suppose

$$W_t = \{2, -1, 0, 3, -2, 1\}$$

Time	1	2	3	4	5	6
W_t	2	-1	0	3	-2	1

Observe that

- Values randomly fluctuate.
- No increasing or decreasing trend.
- No dependence between observations.

Random Walk

A random walk is defined as

$$X_t = X_{t-1} + W_t$$

where W_t is white noise.

Each new observation equals

Previous Value + Random Shock

Hence, random walks possess memory.

Random Walk Example

Suppose

$$X_0 = 100$$

White noise sequence

$$\{2, -1, 3, -2, 1\}$$

$$X_1 = 100 + 2 = 102$$

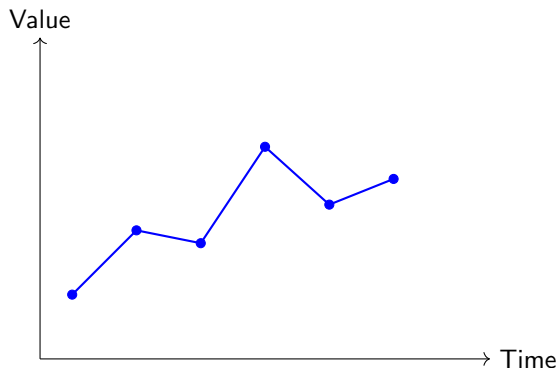
$$X_2 = 102 - 1 = 101$$

$$X_3 = 101 + 3 = 104$$

$$X_4 = 104 - 2 = 102$$

$$X_5 = 102 + 1 = 103$$

Illustration of Random Walk



Unlike white noise, the process accumulates previous random shocks.

Properties of Random Walk

Variance

For white noise,

$$\text{Var}(W_t) = \sigma^2$$

For a random walk,

$$\text{Var}(X_t) = t\sigma^2$$

Hence, uncertainty increases over time.

Properties of Random Walk

Variance

For white noise,

$$\text{Var}(W_t) = \sigma^2$$

For a random walk,

$$\text{Var}(X_t) = t\sigma^2$$

Hence, uncertainty increases over time.

Random Walk with Drift

$$X_t = X_{t-1} + \mu + W_t$$

where

$$\mu$$

is the average change per time period.

Building Blocks: White Noise and Random Walk

White Noise

A collection of uncorrelated random variables w_t with zero mean $\mu = 0$ and finite variance σ_w^2 , denoted $w_t \sim wn(0, \sigma_w^2)$. **Gaussian white noise** additionally assumes w_t are i.i.d. $N(0, \sigma_w^2)$.

Random Walk (with drift δ)

$$X_t = \delta + X_{t-1} + w_t = \delta t + \sum_{i=1}^t w_i$$

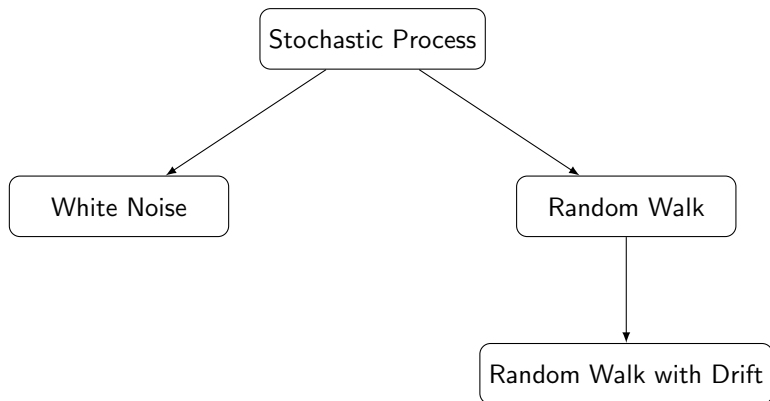
The current value equals the previous value plus a random shock, plus an optional constant drift.

If all time series behavior could be explained by the white noise model, classical statistical methods would suffice. Time series analysis is the study of the (typically much richer) dependence structure beyond white noise.

White Noise vs Random Walk

Property	White Noise	Random Walk
Depends on Previous Value	No	Yes
Mean	Constant	Changes
Variance	Constant	Increasing
Autocorrelation	None	Strong
Stationary	Yes	No
Predictable	No	Short-term Only
Equation	$X_t = W_t$	$X_t = X_{t-1} + W_t$

Relationship Between the Concepts



$$X_t = W_t$$

$$X_t = X_{t-1} + W_t$$

$$X_t = X_{t-1} + \mu + W_t$$

Summary

- A time series is modeled as a stochastic process
- White noise contains only random fluctuations
- White noise is stationary
- Random walk is generated by accumulating white noise
- Random walk is non-stationary because its variance grows with time
- A good forecasting model should produce residuals that resemble white noise

Autocovariance

- Measures the linear relationship of a time series with its **own past values**
- Quantifies the "memory" of the series: How much does today's value depend on the value k steps ago?
- **Crucial for Stationarity:** In a stationary process, autocovariance depends *only* on the lag (k), not on the specific time (t)

Formulation

$$\gamma_k = \text{Cov}(Y_t, Y_{t-k}) = E[(Y_t - \mu)(Y_{t-k} - \mu)]$$

- γ_k : Autocovariance at lag k
- μ : The constant mean of the series.
- **At lag 0 ($k = 0$):** $\Rightarrow \gamma_0 = \text{Cov}(Y_t, Y_t) = E[(Y_t - \mu)^2] = \text{Var}(Y_t)$

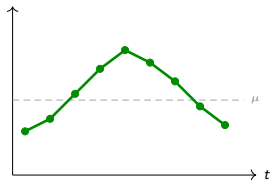
Interpreting the Autocovariance

The autocovariance measures how deviations from the mean at one time relate to deviations at another:

$$\gamma(h) = \text{Cov}(X_t, X_{t+h}) = E[(X_t - \mu)(X_{t+h} - \mu)]$$

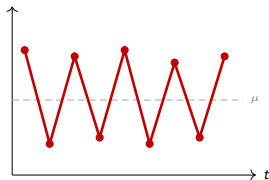
- **Positive:** above average in the past $\Rightarrow$ tends to stay above average now (smooth, persistent).
- **Negative:** above average in the past $\Rightarrow$ tends to be below average now (oscillating, mean-reverting).
- **Zero:** the past carries *no linear* information about the present. With constant mean and variance, this defines **white noise**.

Positive $\gamma(h) > 0$



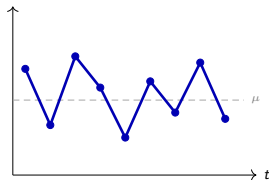
Runs stay on one side of μ : persistent.

Negative $\gamma(h) < 0$



Sign flips each step: mean-reverting.

Zero $\gamma(h) = 0$



No pattern: white noise.

Autocovariance vs. Autocorrelation

Problem with Autocovariance

- Autocovariance: measured in **squared units**
- difficult to interpret the magnitude without knowing the scale of the data

Solution: Autocorrelation (ACF)

- standardize autocovariance by dividing it by the variance (γ_0)
- **Formula:** $\rho_k = \frac{\gamma_k}{\gamma_0}$
- **Autocovariance** (γ_k): Measures the raw, unscaled linear dependence
 - Ranges from $-\infty$ to ∞
- **Autocorrelation** (ρ_k): Measures the scaled, normalized linear dependence
 - bounded between **-1 and +1**
- **Application:** We use the Autocorrelation Function (ACF) plots to identify the p and q parameters in ARIMA models.

Autocorrelation Function (ACF)

$$\rho(s, t) = \frac{\gamma(s, t)}{\sqrt{\gamma(s, s) \gamma(t, t)}}$$

By the Cauchy–Schwarz inequality:

$$\gamma(s, t)^2 \leq \gamma(s, s) \gamma(t, t)$$

so $-1 \leq \rho(s, t) \leq 1$. ACF measures the linear predictability of X_t using only X_s .

Stationarity of a Stochastic Process

Forecasting is difficult: time series is non-deterministic

- But if the series is **stationary**, we can assume its statistical properties (mean, variance, autocorrelation) will be the same in the future as in the past.

Strict Stationarity

- entire probability distribution of the time series is completely invariant (unchanged) when shifted in time
- $\{X_t\}$ is strictly stationary if the joint distribution of $(X_{t_1}, \dots, X_{t_k})$ equals that of $(X_{t_1+h}, \dots, X_{t_k+h})$ for any h
 - joint distribution depends only on the lag h , not on absolute time
- almost impossible to verify in the real world

Weak Stationarity

Weak Stationarity

$\{X_t\}$ is weakly stationary if:

- ① $E[X_t^2] = \sigma^2 < \infty$ for all t (finite variance);
- ② $E[X_t] = \mu$ for all t (constant mean);
- ③ $\gamma_X(s, t) = \gamma_X(s + h, t + h)$ for all s, t, h (autocovariance depends only on lag).

The Relationship

- **Strict $\implies$ Weak:** If the whole distribution is constant (and variance is finite), the mean and variance are constant.
- **Weak $\nRightarrow$ Strict:** Mean and variance can be constant while higher moments (like skewness) change over time.
- **The Gaussian Exception:** If the data is Normally distributed, Weak Stationarity *is* Strict Stationarity.

Weak Stationarity

Weak Stationarity

$\{X_t\}$ is weakly stationary if:

- 1 $E[X_t^2] = \sigma^2 < \infty$ for all t (finite variance);
- 2 $E[X_t] = \mu$ for all t (constant mean);
- 3 $\gamma_X(s, t) = \gamma_X(s + h, t + h)$ for all s, t, h (autocovariance depends only on lag).

Why do we only care about Weak Stationarity?

- models only rely on the mean, variance, and autocovariance (ACF/PACF).
- Weak stationarity provides exactly the mathematical conditions needed to estimate these models.
- Easy to test for using statistical tools (e.g., Augmented Dickey-Fuller test, checking ACF plots).

ACF and PACF for Stationary Series

For a stationary process, $\gamma_X(s, t)$ depends only on $h = s - t$, so we write:

$$\gamma(h) = \text{cov}(X_{t+h}, X_t) = E[(X_{t+h} - \mu)(X_t - \mu)], \quad \rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

Partial Autocorrelation Function (PACF)

The correlation between X_t and X_{t+h} with the linear effect of everything *in between* removed:

$$\phi_{11} = \text{corr}(X_{t+1}, X_t) = \rho_1, \quad \phi_{hh} = \text{corr}(X_{t+h} - \hat{X}_{t+h}, X_t - \hat{X}_t), \quad h \geq 2$$

where $\hat{X}_{t+h}, \hat{X}_t$ are the best linear predictors from the intervening observations. If X_t is Gaussian, ϕ_{hh} is the conditional correlation given $X_{t+1}, \dots, X_{t+h-1}$.

Case Study: Air Passenger Data — Is It Stationary?

Dataset

Monthly totals of US airline passengers, 1949–1960 (a classic time-series benchmark). A time plot reveals a clear **upward trend** and **seasonal fluctuation** — both signs of non-stationarity.

Testing Stationarity: Three Approaches

- **Rolling statistics:** plot rolling mean/variance (e.g., 12-month window); a stationary series shows roughly flat rolling statistics.
- **ACF plot:** non-stationary series often show very slow decay in the ACF.
- **Augmented Dickey–Fuller (ADF) test:** formal hypothesis test. H_0 : the series is non-stationary (has a unit root).

Stationarizing the Series

Since the raw series is non-stationary, we transform it. Common strategies:

- **Detrending:** subtract an estimate of the trend, $\hat{Y}_t = X_t - \hat{\mu}_t$.
- **Differencing:** $\nabla X_t = X_t - X_{t-1}$ (details next section).
- **Transformation:** log, square-root, cube-root, etc., to stabilize variance.
- **Aggregation:** average over a longer period (weekly/monthly).
- **Smoothing:** remove a rolling average from the original series.
- **Decomposition:** explicitly model and remove trend and seasonality.

For the Air Passenger data: apply a **log transform** (to stabilize variance) followed by **first-order differencing** (to remove trend).

The Backshift Operator

Before defining AR/MA/ARMA models concisely, we introduce a key notational tool.

Backshift Operator B

$$BX_t = X_{t-1}, \quad B^k X_t = X_{t-k}$$

extended as $B^2 X_t = B(BX_t) = B(X_{t-1}) = X_{t-2}$, and so on. The inverse (forward-shift) operator satisfies $B^{-1}B = 1$.

Autoregressive (AR) Models

The current value is a linear combination of p past values plus noise:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + w_t = \sum_{i=1}^p \phi_i X_{t-i} + w_t$$

where X_t is stationary, $w_t \sim wn(0, \sigma_w^2)$, and $\phi_p \neq 0$.

Autoregressive Operator

Using the backshift operator:

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p, \quad \phi(B)X_t = w_t$$

AR(1): A Closer Look

$X_t = \phi_1 X_{t-1} + w_t$. Only the previous term and the noise contribute.

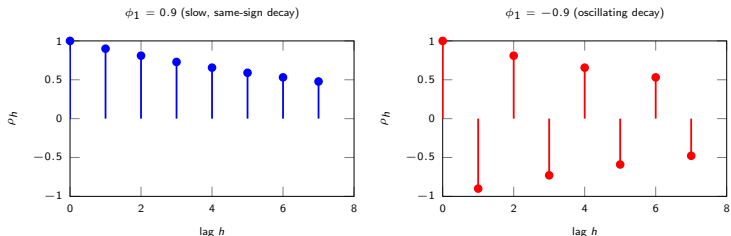
- AR(0) has no dependence — it is essentially white noise.
- If ϕ_1 is close to 0, the process looks like white noise.
- If $\phi_1 < 0$, X_t oscillates between positive and negative values.
- If $\phi_1 = 1$, the process becomes a **random walk** — non-stationary, since the variance grows without bound as $t \rightarrow \infty$.

Mean, Variance, ACF, PACF of AR(1) (with $|\phi_1| < 1$)

$$E[X_t] = 0, \quad \text{Var}(X_t) = \frac{\sigma_w^2}{1 - \phi_1^2}, \quad \rho_h = \phi_1^h$$

$$\phi_{11} = \rho_1 = \phi_1, \quad \phi_{hh} = 0 \quad \forall h \geq 2$$

AR(1): ACF Comparison, $\phi_1 = 0.9$ vs. $\phi_1 = -0.9$



Both decay geometrically as $\rho_h = \phi_1^h$; the sign of ϕ_1 determines whether the ACF stays positive or alternates sign.

Stationarity and the “Stationary Solution” of AR(1)

Iteratively expanding AR(1):

$$X_t = \phi_1 X_{t-1} + w_t = \phi_1(\phi_1 X_{t-2} + w_{t-1}) + w_t = \cdots = \phi_1^k X_{t-k} + \sum_{j=0}^{k-1} \phi_1^j w_{t-j}$$

If $|\phi_1| < 1$ and $\sup_t \text{Var}(X_t) < \infty$, letting $k \rightarrow \infty$:

$$X_t = \sum_{j=0}^{\infty} \phi_1^j w_{t-j} \quad (\text{the **stationary solution**})$$

Moving Average (MA) Models

Motivation

AR models ignore correlated *noise* structure. MA models instead regress the current value on *past forecast errors* $w_{t-1}, \dots, w_{t-q}$, which are informative but unobservable directly.

Definition: MA(q)

$$X_t = w_t + \theta_1 w_{t-1} + \dots + \theta_q w_{t-q} = w_t + \sum_{j=1}^q \theta_j w_{t-j}$$

where $w_t \sim wn(0, \sigma_w^2)$, $\theta_q \neq 0$.

Note

Unlike AR models, a **finite MA model is always stationary** — it is just a weighted moving average of past (finite-variance) shocks.

Moving Average Operator and MA(1) Properties

$$\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q, \quad X_t = \theta(B)w_t$$

$$\text{MA}(1): X_t = w_t + \theta_1 w_{t-1}$$

$$E[X_t] = 0, \quad \text{Var}(X_t) = \sigma_w^2(1 + \theta_1^2)$$

$$\rho_1 = \frac{\theta_1}{1 + \theta_1^2}, \quad \rho_h = 0 \quad \forall h \geq 2$$

General MA(q): Identification via ACF

The theoretical ACF has nonzero values for the first q lags and $\rho_h = 0$ for $h > q$ — the **ACF cuts off after lag q** , making it the natural tool for identifying q (the mirror image of AR identification via PACF).

Definition: ARMA(p, q)

$\{X_t\}$ is ARMA(p, q) if it is stationary and

$$X_t = w_t + \sum_{i=1}^p \phi_i X_{t-i} + \sum_{j=1}^q \theta_j w_{t-j}, \quad \phi_p \neq 0, \theta_q \neq 0, \sigma_w^2 > 0$$

Using the AR and MA operators: $\phi(B)X_t = \theta(B)w_t$.

Identifying (p, q) : ACF/PACF Behavior

	$AR(p)$	$MA(q)$	$ARMA(p, q)$
ACF	Tails off	Cuts off after lag q	Tails off
PACF	Cuts off after lag p	Tails off	Tails off

From ARMA to ARIMA: Differencing

Why Difference?

Many series decompose as $X_t = \mu_t + Y_t$: a non-stationary trend μ_t plus a zero-mean stationary series Y_t . Random walk with drift, $\mu_t = \delta + \mu_{t-1} + w_t$, can represent the trend, giving:

$$\nabla X_t = X_t - X_{t-1} = \delta + w_t + (Y_t - Y_{t-1})$$

the **first difference**. Unlike detrending, differencing requires **no parameter estimation**.

Higher-Order Differencing (via the Backshift Operator)

$$\nabla X_t = (1 - B)X_t, \quad \nabla^2 X_t = (1 - B)^2 X_t = X_t - 2X_{t-1} + X_{t-2}$$

The first difference eliminates a linear trend; the second eliminates a quadratic trend.

The ARIMA(p, d, q) Model

Definition

X_t is ARIMA(p, d, q) if $\nabla^d X_t = (1 - B)^d X_t$ is ARMA(p, q). In general:

$$\phi(B)(1 - B)^d X_t = \theta(B)w_t$$

Parameter	Meaning
p	Lag order — # lagged observations included (AR part)
d	Degree of differencing needed for stationarity (I part)
q	Size of the moving-average window (MA part)

Setting any parameter to 0 drops that component — ARIMA generalizes AR, MA, ARMA, and pure differencing (I) models.

ARIMA(p, d, q)

AR (AutoRegressive) $\rightarrow p$

- **Concept:** Past *values* predict the future.
- Models the "momentum" of the series.
- Regresses Y_t on its own lags ($Y_{t-1}, Y_{t-2}, \dots$).

I (Integrated) $\rightarrow d$

- **Concept:** Removing the trend to achieve **stationarity**.
- Applies *differencing* to the raw data.
- $d = 1$: $Y'_t = Y_t - Y_{t-1}$ (Removes linear trends).
- $d = 0$: Data is already stationary.

MA (Moving Average) $\rightarrow q$

- **Concept:** Past *errors* (shocks) predict the future.
- Models the lingering impact of random surprises.
- Regresses Y_t on past white noise ($\epsilon_{t-1}, \epsilon_{t-2}, \dots$).

Idea

Current value = Past Values (AR) + Past Shocks (MA) + Trend Removal (I).

How to Choose ARIMA Parameters (p, d, q)

Finding the right parameters is the "Goldilocks problem" of time series. We use statistical tools to find the perfect fit.

1. Finding d (Differencing Order)

- **Goal:** Make the data stationary.
- **Method:** Plot the data. If a trend exists, difference it ($d = 1$).
- **Test:** Use the **Augmented Dickey-Fuller (ADF) test** to confirm stationarity.

2. Finding p and q (Lag Orders)

Using Correlograms:

- **PACF plot:** Identifies p . Look for the lag where bars "cut off" to zero.
- **ACF plot:** Identifies q . Look for the lag where bars "cut off" to zero.

Using Information Criteria:

- Use **AIC** or **BIC** to automatically grid-search combinations of p and q .
- Select the model with the **lowest** AIC/BIC score.

Akaike Information Criterion (AIC)

$$AIC = 2k - 2 \ln(\hat{L})$$

where k = number of estimated parameters (e.g., $k = p + q + 1$ for ARMA(p, q) plus variance), and $\hat{L}$ = maximized likelihood of the fitted model.

Corrected AIC (AIC_c) — for Small Samples

$$AIC_c = AIC + \frac{2k(k+1)}{n-k-1}$$

where n = sample size. AIC_c adds a stronger small-sample penalty and converges to AIC as $n \rightarrow \infty$.

Interpretation

Lower AIC is better. The $2k$ term penalizes complexity; $-2 \ln \hat{L}$ rewards fit. Adding a parameter is only worth it if it improves the log-likelihood by more than 1.

Bayesian Information Criterion (BIC)

$$BIC = k \ln(n) - 2 \ln(\hat{L})$$

where n = number of observations.

AIC vs. BIC: Key Difference

BIC replaces AIC's $2k$ penalty with $k \ln(n)$. Since $\ln(n) > 2$ for $n \geq 8$, **BIC penalizes model complexity more heavily** than AIC for typical sample sizes — BIC tends to select **more parsimonious (smaller)** models, especially as n grows.

- AIC is asymptotically efficient for prediction (minimizes forecasting error).
- BIC is consistent for identifying the *true* model order, if one exists in the candidate set, as $n \rightarrow \infty$.

Worked Numerical Example: Comparing ARMA Orders via AIC/BIC

Suppose we fit three candidate models to $n = 128$ stationarized observations (the differenced log Air-Passenger series) and record the maximized log-likelihood $\ln \hat{L}$:

Model	k	$\ln \hat{L}$	$AIC = 2k - 2 \ln \hat{L}$	$BIC = k \ln(n) - 2 \ln \hat{L}$
AR(2)	3	-240.1	486.2	494.5
MA(2)	3	-239.8	485.6	493.9
ARMA(2,2)	5	-233.4	476.8	489.4

(using $\ln(128) \approx 4.852$; e.g.,

$BIC_{AR(2)} = 3(4.852) - 2(-240.1) = 14.56 + 480.2 = 494.8 \approx 494.5$ after rounding of $\ln \hat{L}$)

Interpretation

ARMA(2,2) has both the lowest AIC (476.8) and lowest BIC (489.4) despite having more parameters ($k = 5$) — its substantially better log-likelihood outweighs the complexity penalty under both criteria. This matches the RSS comparison from the earlier case study: ARMA(2,2) fits best.

Causation vs. Correlation

Correlation

A statistical association between two variables: when one changes, the other tends to change too. Measured by $\rho \in [-1, 1]$ (e.g., Pearson correlation). **Symmetric:** $\text{corr}(X, Y) = \text{corr}(Y, X)$.

Causation

A relationship in which changes in X *produce* changes in Y through some mechanism. **Asymmetric:** X causes Y does not imply Y causes X .

Note

Correlation does not imply causation. Two variables can be strongly correlated without either causing the other.

Why Correlation Can Mislead: Confounding

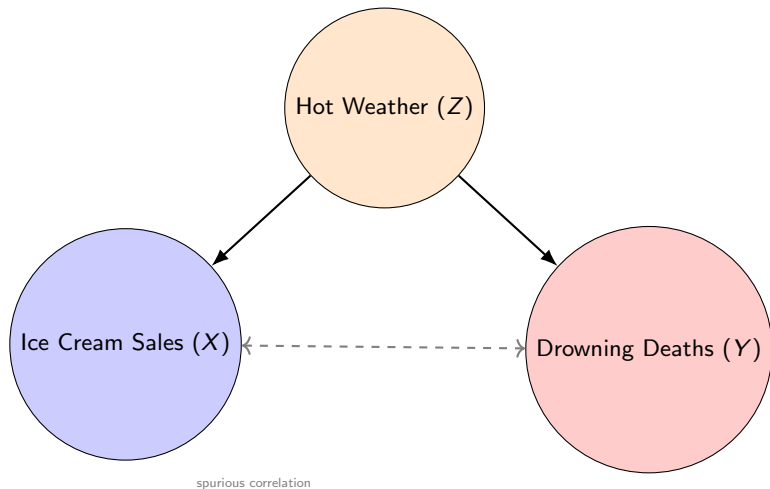
Classic Example

Ice cream sales and drowning deaths are positively correlated. The confounder: **hot weather** increases both ice cream sales and swimming (hence drowning risk). Ice cream does not cause drowning.

Confounding (Common Cause)

A third variable Z causes both X and Y , inducing a spurious correlation between them.

Confounding



The confounder Z (Hot Weather) creates a **backdoor path** $X \leftarrow Z \rightarrow Y$ — producing correlation with **no direct causal arrow** between X and Y .

Granger Causality: Overview

- Developed by Clive Granger (Nobel Laureate, 1969)
- **Core Idea:** If X causes Y , then changes in X must happen *before* changes in Y .
- X "Granger-causes" Y if past values of X help predict Y **above and beyond** what past values of Y alone can predict.
- **Crucial Distinction:** It tests for *predictive information*, not true physical or philosophical cause-and-effect
- A time series X "Granger-causes" Y if **past values of X help predict future values of Y** , beyond what is already predictable from Y 's own past alone

Granger Causality: Formal Setup

Consider two stationary series X_t, Y_t . Compare two regressions for Y_t :

Restricted Model (Y's own past only)

$$Y_t = \alpha_0 + \sum_{i=1}^m \alpha_i Y_{t-i} + \varepsilon_t$$

Unrestricted Model (Y's past + X's past)

$$Y_t = \alpha_0 + \sum_{i=1}^m \alpha_i Y_{t-i} + \sum_{j=1}^m \beta_j X_{t-j} + u_t$$

Test: $H_0 : \beta_1 = \beta_2 = \dots = \beta_m = 0$ (X does not Granger-cause Y) via an **F-test** comparing the residual sums of squares (RSS_r restricted vs. RSS_u unrestricted):

$$F = \frac{(RSS_r - RSS_u)/m}{RSS_u/(n - 2m - 1)} \sim F_{m, n-2m-1} \text{ under } H_0$$

Reject H_0 (conclude X Granger-causes Y) if F exceeds the critical value.

Granger Causality: Assumptions and Limitations

- Both series must be **stationary**: apply the ARIMA
- Sensitive to the choice of lag length $m \rightarrow$ often selected via AIC/BIC
- Cannot detect **instantaneous causality** (effects within the same time period) or causality operating through unobserved confounders.
- **Bidirectional Granger causality** is possible: X may Granger-cause Y and Y may Granger-cause X (feedback systems) $\rightarrow$ test both directions.

Causal Directed Acyclic Graphs (DAGs): Overview

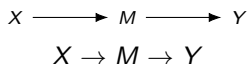
A **causal DAG** represents variables as **nodes** and direct causal effects as **directed edges** ($X \rightarrow Y$ means “ X directly causes Y ”), with **no directed cycles** (you cannot follow arrows and return to where you started).

Why Use a DAG?

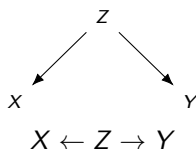
A DAG makes causal assumptions **explicit and visual**, and provides graphical rules (Pearl’s framework) for determining which variables must be statistically controlled for to estimate a causal effect without bias.

Three Basic DAG Structures

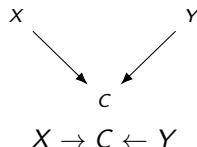
Chain (Mediator)



Fork (Confounder)



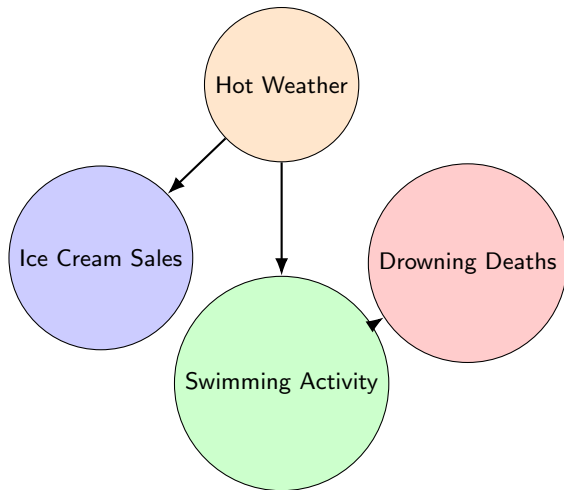
Collider



The Critical Asymmetry

- **Fork:** Z is a confounder — **must control for Z** to get an unbiased estimate of $X \rightarrow Y$.
- **Collider:** C is a collider — **must NOT control for C** ; conditioning on a collider *induces* a spurious association between X and Y .

DAG Example: Revisiting Ice Cream and Drowning



This is a **fork** ($X \leftarrow Z \rightarrow S \rightarrow Y$). To estimate the causal effect of Ice Cream Sales on Drowning, we must **condition on Hot Weather** — this eliminates the spurious association.

Unit V: Summary: Time Series

Autocorrelation and Stationarity

ACF/PACF quantify linear dependence across time; weak stationarity (constant mean, finite variance, lag-dependent autocovariance) is the practical working assumption; the ADF test formally checks for a unit root, demonstrated on the Air Passenger dataset ($0.815 \rightarrow -2.717$ after $\log +$ differencing).

ARMA and ARIMA Models

$AR(p)$ regresses on past values (PACF cuts off at p); $MA(q)$ regresses on past shocks (ACF cuts off at q); ARMA combines both; $ARIMA(p, d, q)$ adds differencing for non-stationary series; Box–Jenkins gives a 3-stage fit-diagnose-refit workflow; SARIMA extends to seasonal data.

AIC/BIC

$AIC = 2k - 2 \ln \hat{L}$, $BIC = k \ln(n) - 2 \ln \hat{L}$ — both trade off fit vs. complexity; BIC penalizes complexity more for $n \geq 8$; used to select (p, q) objectively when ACF/PACF are ambiguous.

Unit V: Summary: Causal Analysis

Causation vs. Correlation

Correlation is symmetric and does not imply causation; confounding, reverse causation, and coincidence are the three main pitfalls; randomized experiments remain the gold standard.

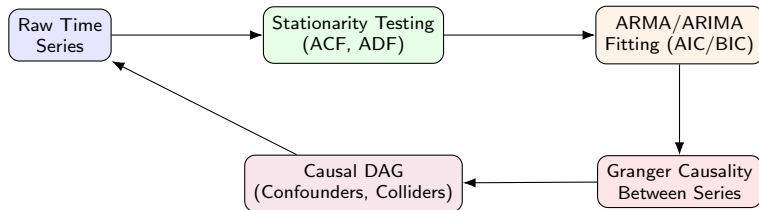
Granger Causality (Overview)

Tests whether past X improves prediction of Y beyond Y 's own past, via an F-test on nested regressions; a statement about predictive precedence, **not** guaranteed true causation.

Causal DAGs (Overview)

Nodes = variables, directed edges = direct causal effects, no cycles; chains (mediators), forks (confounders), and colliders behave very differently under statistical control; the backdoor criterion formalizes which variables to adjust for.

Connecting the Threads



Stationarity is a **prerequisite** for both ARIMA modeling and Granger causality testing. A significant Granger-causality result is still only **predictive**, not confirmatory, evidence of causation - a causal DAG is needed to reason about *why* and to design a valid adjustment strategy.